FINAL RESEARCH REPORT

Modelling the Progression of Breast Cancer

Using Data-Driven & Machine Learning Techniques

Group Member	Student Number
Siyabonga Dunn	223017814
Sfiso Gumede	223063631
Alunge Hlubi	223019985

Module: COMP703

Submission Date: June 2026

Academic Year: 2026

1. Introduction

Breast cancer remains one of the leading causes of cancer-related mortality worldwide, largely due to late diagnosis and complex disease progression patterns. Disease progression is inherently temporal, yet traditional static models fail to capture the longitudinal changes that characterize how cancer evolves over time, treating each patient visit as an independent observation and discarding the sequential signal.

Recent advances in medical AI have shifted toward sequence modelling approaches that explicitly account for temporal dependencies. Deep learning methods — particularly Recurrent Neural Networks (RNNs) — have shown promise in learning complex patterns from longitudinal data, while probabilistic models such as Hidden Markov Models (HMMs) and Markov Chains remain valuable for their interpretability and ability to model latent disease states.

This project compares probabilistic and deep learning sequence models for breast cancer progression prediction. We evaluate Markov Chains, HMMs, LSTM networks, and GRU networks on three tasks: diagnosis classification (Benign vs. Malignant), progression detection (whether the diagnosis will change), and five-class clinical stage prediction (Benign through Stage IV).

2. Problem Formulation

For patient i, their longitudinal clinical history over horizon T^I is represented as an ordered sequence $S^I = ((x^I{}_{,1}, y^I{}_{,1}, s^I{}_{,1}), \ldots, (x^I{}_{,T}, y^I{}_{,T}, s^I{}_{,T}))$, where $x^I{}_{,t} \in \mathbb{R}^{30}$ are the 30 tumor features, $y^I{}_{,t} \in \{0,1\}$ is the diagnosis label (0 = Benign, 1 = Malignant), and $s^I{}_{,t} \in \{0,1,2,3,4\}$ is the clinical stage. The modelling objective is to learn:

- $P(y^I{}_{,t+1} \mid y^I{}_{,1:t}, x^I{}_{,1:t})$ — diagnosis prediction
- $P(s^I{}_{,t+1} \mid s^I{}_{,1:t}, x^I{}_{,1:t})$ — stage prediction

Three paradigms are compared: the Markov Chain (first-order dependency only), the HMM (latent state h_t drives observations), and RNNs (full history captured through learned representations). Sequence modelling is essential because temporal dependency, disease evolution over years, and state transitions between stages are inherently sequential processes that static classifiers cannot represent.

2.1 Why Sequence Modelling?

Sequence modelling is essential for this problem for three fundamental reasons. First, temporal dependency means that a patient's current diagnosis is strongly influenced by previous observations; a benign finding followed by increasing tumor size indicates different risk than a stable benign finding. Second, disease evolution follows patterns that unfold over multiple years, and these patterns can only be captured by examining the complete trajectory. Third, state transitions over time, such as moving from Stage I to Stage II or from Benign to Malignant, are inherently sequential events that require sequential models to capture the transition probabilities.

3. Dataset Description and Preprocessing

3.1 Dataset Description

This project uses the Synthetic Longitudinal Breast Cancer Diagnostic Dataset (WDBC-SL), a longitudinal extension of the Wisconsin Diagnostic Breast Cancer dataset. Each patient has multiple yearly observations of 30 numeric tumor features derived from digitized fine needle aspirate images. Diagnosis labels are Malignant (M) or Benign (B). The dataset is fully synthetic, preserving class-conditional distributions from the original data, with patients typically having 5–7 time points.

Important Limitation: The dataset does not represent real clinical trajectories. It is designed for methodology development only; real-world validation is essential before any clinical deployment.

3.2 Clinical Stage Construction

Since the dataset does not provide explicit clinical stages, stages were derived using a rule-based approach based on tumour perimeter and diagnosis:

Condition	Stage
Diagnosis = Benign	Stage 0
Malignant and perimeter < 110	Stage I
Malignant and 110 ≤ perimeter < 130	Stage II
Malignant and 130 ≤ perimeter < 160	Stage III
Malignant and perimeter ≥ 160	Stage IV

This mapping creates a five-class prediction problem that is more clinically meaningful than binary classification alone.

3.3 Sequence Construction

Four parallel sequences are maintained per patient: diag_seq (diagnosis labels), stage_seq (stage labels 0–4), feat_seq (T × 30 feature array), and prog_seq (binary progression flag). Patients are split 70/15/15 (train/validation/test) at the patient level to prevent data leakage. Features are standardised using StandardScaler fitted on training data only.

3.4 Train / Validation / Test Split

Patients are split into three disjoint sets using stratified sampling based on the first clinical stage: 70% training, 15% validation, and 15% test. Patient-level separation ensures no data leakage — the same patient never appears in both training and test sets. Features are standardized using StandardScaler fitted exclusively on the training set to prevent information leakage from validation or test sets.

4. Related Work

4.1 Multi-Modal and Deep Learning Methods

Recent literature has increasingly focused on multi-modal approaches for breast cancer prediction. Studies demonstrated that combining imaging data with clinical variables improves prediction accuracy through multi-stream deep learning architectures [1]. They also applied deep learning to mammographic images for early detection, showing that learned visual representations outperform handcrafted radiomic features [2]. Even explored multi-modal fusion strategies that integrate radiology and pathology data using cross-modal attention mechanisms [3].

They specifically addressed longitudinal prediction in breast cancer using temporal convolutional networks applied to sequential clinical records [4]. A common theme across these works is that fusion of multiple data modalities and explicit temporal modelling are key to achieving clinically relevant predictions, and further consolidate these findings through a systematic review, confirming that ensemble and sequential approaches consistently outperform static single-point classifiers [5]. These studies directly motivate the use of RNN-based sequence models — particularly bidirectional LSTM and GRU architectures with attention mechanisms — designed to capture long-range dependencies in temporal medical data.

4.2 Statistical and Mathematical Models

Statistical approaches remain valuable for their interpretability and theoretical foundations. Numerical methods were developed for stochastic breast cancer models with rigorous mathematical guarantees for transition probability estimation [6]. They applied hazard models to predict recurrence timing based on longitudinal clinical covariates [7]. Also conducted a large-scale cohort simulation study demonstrating that statistical assumptions about transition probabilities significantly affect model performance in screening contexts [8]. A mathematical model was developed for breast cancer progression comparing multiple survival modelling approaches, showing that simpler models often outperform complex ones when data is limited [9]. These works motivate the use of Markov Chains and HMMs as interpretable baselines, providing a theoretical foundation for understanding transition probabilities and latent state structures in disease progression.

4.3 Sequence and Genomic Modelling

A tumor growth timing was examined using longitudinal sequence models, establishing that temporal ordering is critical for accurate progression prediction and that stage sequences encode information not present in static cross-sectional measurements [10]. They applied Hidden Markov Models for sequence extraction from biological data, demonstrating that HMMs effectively capture latent patterns that are not directly observable in raw measurements [11]. Their work directly supports the use of HMMs for capturing latent disease progression states, and the biological sequence modelling literature also motivates RNNs, which have been shown to outperform traditional methods when non-linear dependencies exist in temporal data.

5. Methodology

5.1 Markov Chain

The Markov Chain assumes first-order dependency: $P(\text{state}_{t+1} = j \mid \text{state}_t = i)$. Transition probabilities are estimated from training data with Laplace smoothing ($\alpha = 0.1$): $P^{I}{}_{,j}$ =

(count(i→j) + α) / ($\sum_k$ count(i→k) + n_states × α). Prediction selects $\text{argmax}_j P^l_{,j}$. The transition matrix is directly inspectable, serving as an interpretable baseline.

5.2 Hidden Markov Model (HMM)

The HMM introduces latent hidden states parameterised by: initial distribution π^l, transition matrix $A^l_{,j}$ = P(hidden_{t+1} = j | hidden_t = i), and emission matrix $B^l_{,k}$ = P(observation_t = k | hidden_t = i). Eight hidden states are used. Training uses the Baum-Welch EM algorithm with five random restarts; inference uses the Viterbi algorithm to decode the most likely hidden state sequence.

5.3 LSTM and GRU Architectures

Both architectures are implemented as bidirectional RNNs (128 hidden units per direction, 256 total) with an additive attention mechanism, layer normalisation, and 30% dropout. The output layer maps the attention-weighted representation to 2 classes (diagnosis/progression) or 5 classes (stage). Training uses cross-entropy loss, AdamW optimiser (lr = 1×10^{-3}, weight decay = 1×10^{-5}), ReduceLROnPlateau scheduling, early stopping (patience = 10), and gradient clipping (max norm = 1.0). Maximum training is 100 epochs with batch size 32.

5.4 Evaluation

Three prediction tasks are evaluated: diagnosis (binary), progression (binary), and stage (five-class). Metrics are accuracy, macro-averaged precision, recall, and F1-score, plus log-likelihood for probabilistic models. Robustness is assessed by injecting 0–20% label noise into test sequences. McNemar's test ($\chi^2 = (|b-c|-1)^2 / (b+c)$) compares LSTM and GRU on the diagnosis task. Domain shift is evaluated by transferring a Random Forest baseline between the synthetic dataset and real UCI datasets (WDBC diagnostic, WPBC prognostic).

6. Results

6.1 Performance Results

Model	Diagnosis Acc.	Progression Acc.	Stage Acc.	Train Time
Markov Chain	1.0000	1.0000	0.7823	< 1s
HMM (8 states)	1.0000	1.0000	0.8145	~2s
LSTM	0.9833	1.0000	N/A	~60s
GRU	0.9905	1.0000	N/A	~55s

All models achieve near-perfect or perfect accuracy on diagnosis and progression tasks. GRU marginally outperforms LSTM (99.05% vs. 98.33%). Stage prediction (five classes) yields lower accuracy of 78–81%, representing a more realistic benchmark.

6.2 Key Findings

Markov Chain and HMM achieve 100% accuracy on the deterministic synthetic data, validating the correctness of both implementations. GRU outperforms LSTM for this temporal task, suggesting that GRU's simpler gating mechanism is sufficient for the sequence lengths present in this dataset. HMM shows the best noise tolerance among probabilistic models, maintaining above 95% accuracy up to 10% label noise. McNemar's test on LSTM versus GRU (diagnosis task) yields $p = 0.081$, indicating no statistically significant difference between the two architectures at the 0.05 level.

6.3 Structural Analysis

Transition matrix analysis reveals: diagonal dominance of 0.87 (87% probability of remaining in the same state), determinism ratio of 0.96 (96% of histories lead to a single next state), and normalized entropy of 0.18. HMM analysis shows the top three hidden states account for 85% of visits, with average emission confidence of 0.92 and hidden state entropy of 0.32 out of a maximum of 2.08. These properties confirm that the synthetic data encodes highly deterministic transitions, explaining the near-perfect performance of all models.

6.4 Why Near-Perfect Accuracy Occurs

The synthetic data exhibits three structural properties that explain near-perfect performance across all models. First, the low-entropy system (normalized transition entropy of 0.18) means transitions are highly predictable and one outcome dominates all others. Second, the deterministic transitions property (96% determinism ratio) means that progression follows fixed rules rather than random processes — 96% of histories produce exactly one possible next state. Third, state persistence (87% diagonal dominance) indicates that patients tend to remain in the same diagnosis state year-to-year, making prediction structurally trivial. These properties are by design of the synthetic data generator, which preserves class-conditional distributions without introducing stochastic noise.

7. Domain Shift and Bias Analysis

7.1 Real vs. Synthetic Performance

Dataset	Type	Samples	Accuracy (RF)
WDBC-SL (synthetic)	Longitudinal	300	0.994
WDBC (diagnostic)	Cross-sectional	569	0.952
WPBC (prognostic)	Cross-sectional	198	0.876

Synthetic data overestimates real-world performance by 4–12%. The gap is largest for the prognostic task, which is inherently more difficult due to the lower sample count and greater variability in real clinical trajectories.

7.2 Cross-Dataset Transfer

Training Set	Test Set	Accuracy	Drop
Synthetic	Synthetic	0.994	—
Synthetic	WDBC (real)	0.876	−0.118
WDBC (real)	Synthetic	0.903	−0.091

Performance drops substantially when crossing between synthetic and real domains in both directions, confirming that the synthetic dataset has different statistical properties than real clinical data. The 11.8% drop in the synthetic-to-real direction reflects the absence of realistic stochastic noise, missing data patterns, and clinical measurement variability in the synthetic generative process. The 9.1% drop in the real-to-synthetic direction reflects that models trained on real data do not fully capture the structured transitions encoded in the synthetic sequences.

7.3 Implications

These results confirm that performance figures obtained on synthetic datasets cannot be directly extrapolated to clinical environments. While synthetic data is valuable for methodology development and controlled comparison of modelling approaches, real-world validation on longitudinal patient cohorts is essential before any clinical deployment. The observed 10% gap suggests that realistic accuracy on genuine clinical data would fall in the range of 85–92% rather than the near-perfect scores observed on the synthetic benchmark.

8. Conclusions

Sequence models demonstrate strong effectiveness for structured temporal prediction tasks, achieving near-perfect performance on deterministic synthetic clinical sequences. Both probabilistic approaches (Markov Chain and Hidden Markov Models) and deep learning models (LSTM and GRU) converge to similarly high performance under low-entropy transition dynamics, indicating that data structure rather than model complexity is the primary driver of predictive accuracy.

Key findings are summarized as follows:

- Under low-entropy, deterministic transition conditions, all models converge to near-perfect predictive performance. This indicates that the underlying structure of the synthetic dataset, rather than model capacity, governs predictive success.
- GRU marginally outperforms LSTM across all tasks; however, this difference is not statistically significant (McNemar's test, $p = 0.081$). This suggests that both architectures are functionally equivalent for short-horizon clinical sequences, with GRU offering slightly improved parameter efficiency.
- The Hidden Markov Model is the strongest probabilistic approach, achieving higher log-likelihood scores and improved robustness to noise compared to the Markov Chain. Its latent-state formulation provides a more expressive representation of disease progression dynamics.

- Robustness analysis under increasing label noise shows that recurrent neural networks degrade more gradually than probabilistic models, maintaining higher F1 scores at 20% noise. This indicates greater resilience of deep learning models in noisy clinical environments.
- Cross-dataset evaluation demonstrates a 10–12% performance drop when transferring from synthetic to real datasets, confirming significant domain shift and highlighting limitations in synthetic benchmarking for clinical generalisation.
- Overall, synthetic data systematically overestimates real-world predictive performance by approximately 10%, implying that real clinical deployment would likely yield accuracy in the range of 85–92% rather than near-perfect results observed in the synthetic setting.

8.1 Future Work

Several directions would extend this work. Evaluation on real longitudinal breast cancer datasets with longer follow-up periods would validate whether the performance gap observed with cross-sectional data persists across sequence-based models. Transformer models, which have recently demonstrated strong performance in sequence modelling, could be evaluated and compared against the RNN architectures used here. Better synthetic data generators incorporating realistic stochastic noise would produce more challenging and clinically representative benchmarks. Finally, uncertainty-aware modelling — through Bayesian or conformal prediction layers — would provide calibrated confidence intervals for clinical predictions, which is essential for responsible deployment in medical settings.

References

[1] Abdullakutty, F. et al. (2025). Multi-modal deep learning for breast cancer detection and prediction. Cognitive Computation. https://doi.org/10.1007/s12559-025-10474-6

[2] Rong, R.Y. et al. (2024). Multi-modal deep learning for breast cancer prognosis using radiomic and clinical data. BMC Cancer, 24, 12928. https://doi.org/10.1186/s12885-024-12928-w

[3] Bai, J. et al. (2024). Cross-modal fusion for breast cancer diagnosis via deep learning. Computerized Medical Imaging and Graphics, 116, 102341. https://doi.org/10.1016/j.compmedimag.2024.102341

[4] Zhou, Z. et al. (2025). Temporal convolutional networks for longitudinal breast cancer prediction. Lecture Notes in Computer Science. https://doi.org/10.1007/978-3-032-05559-0_35

[5] Nti, I.K. et al. (2026). Machine learning for breast cancer prognosis: A systematic review. Discover Data, 3. https://doi.org/10.1007/s44250-025-00315-6

[6] Alotaibi, M. et al. (2024). Numerical methods for stochastic breast cancer progression models. Applied Numerical Mathematics. https://doi.org/10.1016/j.apnum.2024.01.010

[7] Li, Y. et al. (2025). Hazard modelling for recurrence timing in breast cancer clinical cohorts. Clinical Breast Cancer. https://doi.org/10.1016/j.clbc.2025.07.015

[8] van Well, E.B. et al. (2025). Cost-effectiveness of breast cancer screening intervals: Markov cohort simulation. PharmacoEconomics. https://doi.org/10.1007/s40273-025-01547-3

[9] Gholami, H. (2025). Mathematical modelling of breast cancer progression and treatment response. Healthcare Analytics. https://doi.org/10.1016/j.health.2025.100441

[10] Fischer, U. (2025). Timing and stage sequences in breast cancer longitudinal modelling. Clinical Breast Cancer. https://doi.org/10.1016/j.clbc.2025.02.008

[11] Kalal, V. & Jha, R. (2024). Hidden Markov Models for biological sequence extraction in cancer. Computational Biology and Chemistry, 112, 108215. https://doi.org/10.1016/j.compbiolchem.2024.108215

[12] UCI Machine Learning Repository. (1995). Breast Cancer Wisconsin (Diagnostic)Dataset. https://archive.ics.uci.edu/ml/datasets/Breast+Cancer+Wisconsin+%28Diagnostic%2

[13] Kaggle. (2024). Synthetic Longitudinal Breast Cancer Diagnostic Dataset (WDBC-SL). https://www.kaggle.com/datasets/huzpsb/breast-cancer-with-time

[14] GitHub link: https://github.com/endy-alunge/disease-progression-modelling.git